

InfoVis Exoplanets — Project Documentation

Interactive Information Visualization · SS26 · Technische Hochschule Nürnberg

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Link to the project repository: <https://github.com/Laurin-K/InfoVis-Exoplanets>

Link to the deployed web application: <https://laurin-k.github.io/InfoVis-Exoplanets/>

1. Project Overview

This project is an interactive web-based visualization tool for exploring the [NASA Exoplanet Archive](#) dataset. The application enables the users to explore thousands of exoplanets through multiple coordinated and standalone views built with **D3.js**. The target user groups for this visualization are public audiences and students.

The tool is structured around three functional modules:

Module	Views
Main Views	Parallel Coordinate Plot, Spiderplot, Spiderplot Gallery, Scatterplot
Interactive Labs	Habitable Zone Explorer, Gravity Drop Simulator, Transit Light Curve Lab
Utilities	Planet Selector

2. Final Design

2.1 Application Architecture

```
InfoVis-Exoplanets/  
├── index.html           # Landing page  
├── pages/               # HTML pages for every visualization  
├── scripts/             # D3.js visualization logic (JS)  
├── styling/             # CSS per page + shared base  
└── data/               # NASA CSV exports + Python processing scripts
```

2.2 Technology Stack

- **Frontend:** HTML5, CSS, JavaScript

- **Visualization:** [D3.js v7](#)
- **Data Source:** NASA Exoplanet Archive (public API + CSV export)
- **Data Processing:** Python

2.3 Main Views

Parallel Coordinate Plot ([parallel-coordinate-plot.html](#))

Displays multiple numeric dimensions as parallel vertical axes. Users are able to see trends or spot outliers for every two dimensions.

Key functionality:

- **Axis Selection:** Users can select which dimensions to show/hide via a control panel
- **Brushing:** Filtering method for each axis to narrow down the results; Lines outside the filter range are greyed out
- **Toggle reference planet:** Earth ↔ Jupiter unit toggle for mass and radius
- **Axis Scaling:** Axis are scaled based on user preference (Auto / Linear / Log), with per-axis LOG/LIN labels
- **Coloring** LCh equal-lightness color palettes
- **Hovering:** Highlights a single line; click selects the planet and shows an infocard
- **Missing Data:** Displayed in the chart with dashed lines; serves as line continuity

Quality of life:

- Fullscreen chart toggle
- Reset buttons for all filters or per axis
- Glossary table with column names and explanations

Scatterplot ([scatterplot.html](#))

Allows the comparison of 3 dimensions (including coloring). Users are able to see clusters or emerging patterns.

Key functionality:

- **Axis selection:** Users are able to select, change and flip axes
- **Pan & Zoom** Panning and zooming interaction modes
- **Overview & Detail:** Zoom + Overview window; The user is able to zoom into the scatterplot without losing orientation
- **Coloring:** Users are able to select different coloring options
- **Solar System overlay:** Adds all planets from the earth's solar system as reference points
- **Discovery Year slider** with Play animation (planets appear chronologically starting at 1992 up to 2026)

Quality of life:

- Reset Zoom button
- Planet infocard on click (with link to exoplanet archive)
- Data fields & definitions table at the bottom

Spiderplot ([spiderplot.html](#))

Allows the comparison of 4 planets for up to 8 dimensions in a radial view. Users are able to compare planets from the same planet system

Key functionality:

- **Axis selection:** Users are able to add or remove axis for better comparison
- **Search:** Searching for planets by name, with live dropdown preview; can be limited to planets from
- **Hover:** Hovering over one axis shows values with units
- **Visibility toggle:** per planet in the legend
- **Missing data:** "No Data" placeholder per axis (dashed lines)

Quality of life:

- **Axes values visibility:** toggle to always on/off
- **Reposition axis:** order of the axes can be changes for specific needs
- **Adjust scaling:** either all log or all lin, depending on the users need

Spiderplot Gallery ([spiderplot-gallery.html](#))

Grid view of individual spiderplots providing an overview of all planets. Selecting one planet allows the user to dive deeper into the planets data with a familiar spider plot view. Sorting is based on the planet system as mentioned in the expert feedback.

2.4 Additional views

Habitable Zone Explorer ([habitable-zone.html](#))

- Visualizes habitable zone boundaries based on stellar properties
- Selection by planetary system (not individual planet)
- Shows which planets in a system fall within/outside the habitable zone

Gravity Drop Simulator ([gravity-drop.html](#))

- Simulates dropping an object on different exoplanet surfaces
- Compares surface gravity across selected planets
- Playful/educational: shows drop time relative to Earth

Transit Light Curve Lab ([transit-lab.html](#))

- Simulates the photometric dip as a planet transits its host star
- Y-axis is fixed; multiple transit profiles are overlaid for comparison
- Educational explanation of the transit detection method

Planet Selector ([planet-selector.html](#))

- Utility tool to build a custom planet dataset for the other visualizations
- Filter and select planets by properties

3. Process

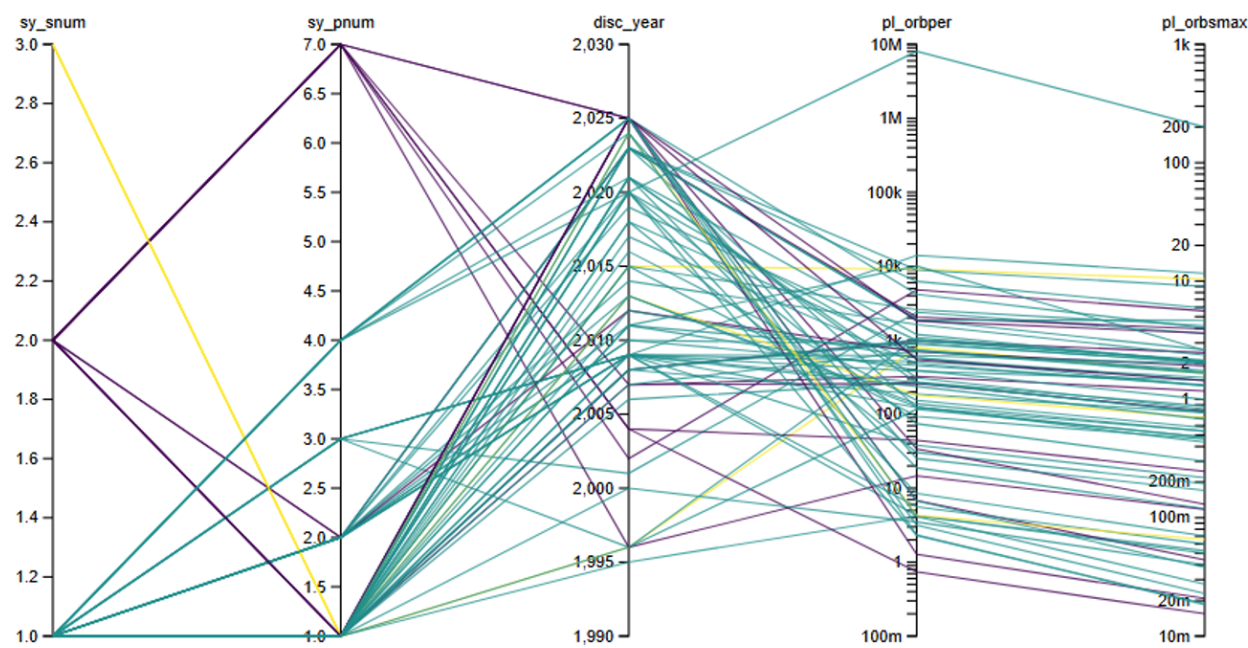
3.1 Initial Ideas, Sketches and Prototyping

At this stage, the following core views were proposed:

- Parallel Coordinate Plot (main multi-dimensional exploration tool)
- Spiderplot for individual planet comparisons
- Scatterplot for two-variable relationships
- Scatterplot Matrix (discussed but not pursued)
- Multiple Coordinated Views (discussed, evaluated)

Prototype:

- First Application prototype with a simple Parallel-Coordinate-Plot:
- Small sample of the NASA Exoplanet Archive



Key open questions at Week 6:

- What to do with incomplete/missing data rows?
- Who is the target audience? Researchers or students?
- Should a Scatterplot Matrix be added?
- Should a Spiderplot be added?
- Should Principal Component Analysis (PCA) be added as a dimension-reduction toggle?
- What is the design language (styling) going to be?

3.2 Early Implementation Milestones

Date	Development stage	Implemented functionality

Date	Development stage	Implemented functionality
2026-05-30	Initial prototype (PCP)	A first Parallel Coordinate Plot was implemented in a single HTML file using D3.js. It displayed five dimensions from a small NASA Exoplanet Archive dataset.
2026-05-31	Application structure	The project was divided into a landing page, visualization pages, shared styling and a data directory. Placeholder pages for the Scatterplot and Spiderplot were introduced. The structure was subsequently adapted for deployment through GitHub Pages.
2026-05-31	Automatic axis scaling	The Parallel Coordinate Plot received automatic scale selection. Dimensions spanning more than two orders of magnitude were displayed using logarithmic scales; other dimensions used linear scales.
2026-05-31	Configurable PCP axes	Users could add or remove dimensions through dynamically generated checkboxes. The visualization was redrawn whenever the selection changed.
2026-06-04	PCP reference units and refactoring	Users could switch planetary mass and radius between Earth-based and Jupiter-based units. The visualization logic and styling were moved from the HTML file into separate JavaScript and CSS files. A glossary generated from column_explanation.csv was added.
2026-06-04	Multi-axis brushing	Vertical D3 brushes were added to every PCP axis. Multiple brush ranges could be combined to filter planets across several dimensions.
2026-06-04	First Scatterplot	A standalone Scatterplot Explorer was introduced. It supported selectable and swappable X/Y dimensions, automatic linear or logarithmic scaling, discovery-year coloring, hover tooltips and a data glossary.
2026-06-04	First Spiderplot	The first Spiderplot comparison view supported up to four planets and eight normalized metrics. It included planet search, initial example planets and a separate searchable gallery of individual spiderplots.
2026-06-05	PCP interaction improvements	Tooltips for axis labels were added using the column glossary. Lines became clickable, allowing users to highlight a planet and display an information card.
2026-06-07	Unified visual design	The landing page and Parallel Coordinate Plot received a shared visual language and more consistent page, control and chart styling.

These milestones formed the first functional version of the application.

3.3 Feedback Round 1 - 2026-06-10

Source: [feedback/20260610-Feedback.md](#)

Topic	Feedback
General	"Missing insights": The app needs clearer takeaways
General	Expert consultation required

Topic	Feedback
General	Incomplete data: show dashed lines for partial entries
PCP	Fullscreen toggle for the chart
PCP	PCA toggle option
Scatterplot Matrix	Add it; click to focus
Spiderplot	Better handling of NULL values; selectable dimensions
Spiderplot Gallery	Show axis labels/values; selectable dimensions; filter; clustering; selection/overlap comparison
Multiple Coordinated Views	Evaluate whether to add

3.4 Feedback Round 2 - 2026-06-24

Source: [feedback/20260624-Feedback.md](#)

App purpose clarification: Should the app focus on exploratory or expert work?

User Study: More intuitive operation for each plot; better visibility of toggles and selections.

Topic	Feedback	Implemented?
General	Add tutorial / help button	Yes
PCP	Movable axes	Yes
PCP	Make Earth/Jupiter toggle more visible	Yes
PCP	Larger brush hit area + Reset brushing	Yes
PCP	Tooltip for dimension names	Yes
PCP	Fullscreen toggle	Yes
Spiderplot	Live search preview in dropdown	Yes
Spiderplot	Add planet directly on select	Yes
Spiderplot	Hover tooltip with values	Yes
Spiderplot	Eye-icon hide/show per planet	Yes
Spiderplot	"No Data" per quadrant	Yes
Scatterplot	Zoom into plot	Yes
Scatterplot	Panning	Yes
Scatterplot	More color options	Yes

3.5 Expert Feedback - 2026-06-31 (Laura von Zadow)

Source: [feedback/20260631-Feedback-Experte.md](#)

Key expert insight: Target audience is public outreach (students/enthusiasts), not researchers.

Topic	Expert Feedback	Implemented?
General	Link planets to NASA exoplanet archive	Yes
General	"Multi-planet systems" visualization (system lines)	No
Spiderplot	Show units on axis labels, not just on hover	Yes
Spiderplot	Disclaimer for missing data (with fallback to secondary dataset)	Yes
Spiderplot	Dimensions add/remove	Yes
Spiderplot	Preload 2–4 demo planets	Yes
Scatterplot	Solar System planets as reference points	Yes
Scatterplot	Default: Orbital Period (days) vs. Jupiter Radius	Yes
Scatterplot	Discovery Year animation/timeline	Yes
Scatterplot	Filter/color by Discovery Method	Yes
Scatterplot	Click → planet infocard + link	Yes
Scatterplot	Performance improvements	Yes
PCP	LCh color space for equal-lightness palettes	Yes
PCP	Log/Linear labels per axis	Yes
PCP	Interesting default dataset	Yes
Habitable Zone	Only show systems with multiple planets (orbit period ≥ 10 days)	Yes
Gravity Drop	Add to application, for public outreach	Yes
Transit	Stable Y-axis; show how drop changes	Yes

3.6 Feedback Round 3 - 2026-07-01

Source: [feedback/20260701-Feedback.md](#)

Topic	Feedback	Implemented?
General	Check all axis labels	Yes
General	Add Solar System planets for context	Yes
PCP	Hover → better highlighting for one line, fade others	Yes
PCP	Poor planet visibility when all lines are plotted	Yes
PCP	LCh color space	Yes
PCP	Log/Linear labels + toggle	Yes

Topic	Feedback	Implemented?
Spiderplot	Re-add example planets	Yes
Spiderplot	Axes start fixed at 0	Yes
Spiderplot	Avoid mixing log/linear scales	Yes

3.7 Feedback Implementation

Based on several feedback rounds with users and experts, the application was further refined. The collected comments and requirements were carefully evaluated and incorporated into the final implementation.

4. Design Rationale & Alternatives

4.1 Parallel Coordinate Plot

The NASA dataset has 30+ numeric columns. PCP lets users see patterns across all dimensions simultaneously and filter interactively via brushing. Hard to achieve in a standard table.

Alternative considered: Spiderplot. Discussed in several feedbacks. Implemented as an alternative view.

4.2 Spiderplot

Ideal for comparing a small number of planets across normalized metrics at a glance. Complements the PCP by focusing on individual planet profiles rather than trends.

4.3 Scatterplot

Ideal plot for finding clusters in the data. Two modes (Pan and Box Zoom) let users navigate freely and also zoom precisely into dense clusters without losing context.

4.4 LCh Color Space

Standard HSL palettes produce colors of varying perceived brightness (purple appears much darker than yellow at the same lightness value). LCh (Lightness-Chroma-Hue) keeps all hues perceptually equal-brightness, making every line visible regardless of value. This was directly recommended by the expert and confirmed in the 2026-07-01 feedback.

5. Who Did What

Decisions were made collectively by the team. Each view head was responsible for the oversight of the development and changes made by him or the team. The development process was executed through the utilization of pair or team programming methodologies. Every member of the team contributed equally to the success of the project.

Member	Contributions
Laurin Kerntke	Head of Parallel Coordinate Plot View

Member	Contributions
Jonathan Kechter	Head of Scatter Plot View, Planet selector, Interactive Labs
Daniel Schweiger	Head of Spider Plot View

6. Data

File	Description
<code>nasa_export_large.csv</code>	Full NASA Exoplanet Archive export (~900 KB)
<code>nasa_export_small.csv</code>	Reduced export for fast loading (~15 KB)
<code>nasa_export_large_merged.csv</code>	Large export merged with secondary API data
<code>nasa_export_small_merged.csv</code>	Small export merged with secondary API data
<code>all_exoplanets_2021.csv</code>	Historical snapshot dataset (2021)
<code>api_only_export.csv</code>	Raw API export
<code>column_explanation.csv</code>	Human-readable descriptions of all data columns
<code>fetch_api_data.py</code>	Python script to fetch data from the NASA API
<code>merge_datasets.py</code>	Python script to merge/clean the datasets
<code>spiderplot-data.js</code>	Pre-processed JS data file for the spiderplot (~880 KB)

7. Limitations

Items that were requested in feedback but not yet confirmed as implemented:

- ☐ Spiderplot Gallery: clarify purpose, might not get used as intended or serve a meaningful purpose
- ☐ PCP: improve single-planet visibility when all lines are rendered, handling of overlapping lines
- ☐ Scatterplot: performance improvements for large datasets
- ☐ "System comparison" visualization: Expert feedback: one line per system, planets on the line, user is able to see planets distances to each other and can compare those to other systems